DS Environmental CONSULTING

"The trusted choice for your environmental & industrial hygiene needs."

ASBESTOS INSPECTION REPORT

275 Holbrook St, Erie, Colorado

PRESENTED TO:	INSPECTED BY:	PROJECT DETAILS:
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1.0 Introduction

Mr. Chris Lehman with DS Environmental Consulting (DS) conducted an asbestos inspection and collected bulk-samples of suspect asbestos-containing building materials within the single-family residence detailed on the cover page of this report. The purpose of the inspection was to determine if any of the materials that may be impacted by renovation activities contain asbestos.

2.0 Limitations of Inspection

This inspection was limited in its scope and only included specific areas and materials as defined by the client. The inspection did not constitute a full-building inspection and does not fulfill the asbestos inspection requirements for structures that are to be demolished.

The table below, (*Table 1.0*), lists the suspect asbestos-containing materials included in the scope of the inspection. It identifies the specific areas that were included in the inspection as well descriptions of the suspect asbestos-containing materials in those areas that were sampled; or materials that were assumed to contain asbestos.

The suspect asbestos-containing materials listed below may be found in other areas of the building that were outside of the scope of work for this inspection and sampling. Additionally, there may be other materials in other parts of the building that contain asbestos, which were not included in the scope of this inspection and sampling.

Table 1.0	Sampled or Assumed Suspect ACM within Scope of Work

Materials in **RED** are materials that contain greater than 1% asbestos.
Materials in **BLUE** are assumed to contain >1% asbestos
Materials in **GREEN** contain 1% asbestos or less.
Materials in **BLACK** are none-detected for asbestos.

Suspect Asbestos-Containing Materials Sampled		Material Locations **See Appendix B for Sample Location Map
Homogeneous Area 1 (SM1)	Smooth Nap Textured Drywall	· Kitchen/laundry, east entry/mud room, SE (primary) bedroom and associated closet, NE and NW bedroom closets, hallway and bathroom walls and ceilings · NW bedroom ceiling
Homogeneous Area 2 (JC1)	Joint Compound Associated w/ SM1	· Kitchen/laundry, east entry/mud room, SE (primary) bedroom and associated closet, NE and NW bedroom closets, hallway and bathroom walls and ceilings
Homogeneous Area 3 (SM2)	Sponge Stomp Textured Drywall	· NE bedroom and hallway walls and ceilings · Living room, north and south walls · West entry foyer ceiling
Homogeneous Area 4 (JC2)	Joint Compound Associated w/ SM2	· NE bedroom and hallway walls and ceilings · Living room, north and south walls · West entry foyer ceiling

Homogeneous Area 5 (SM3)	Heavy Knock Down Textured Drywall	· West entry foyer walls
Homogeneous Area 6 (JC3)	Joint Compound Associated w/ SM3	· West entry foyer walls
Homogeneous Area 7 (SM4)	Swirl Textured Fiberboard	· Living room ceiling under ceiling tiles (CT1) and associated wood framing
Homogeneous Area 8 (JC4)	Joint Compound Associated w/ SM4	· Living room ceiling under ceiling tiles (CT1) and associated wood framing
Homogeneous Area 9 (CT1)	1'x 1' White Ceiling Tiles with Flower and Dot Design	· Living room ceiling over fiberboard (SM4/JC4) and associated wood framing
Homogeneous Area 10 (INS1)	Grey/Beige Cellulose Insulation	· Attic
Homogeneous Area 11 (FT1)	12" White/Black Vinyl Floor Tiles and Associated Yellow Adhesive	· Kitchen floor
Homogeneous Area 12 (ECLK1)	White Exterior Caulking	· Exterior windows

3.0 Conclusions & Summary of Findings

	ACRONYMS	ACM ASSESSMENT CATEGORIES
SUMMARY OF FINDINGS	CHRY – Chrysotile ACT – Actinolite TR – Trace; Assumed >1% Asbestos ND – None-detected ACM – Asbestos Containing Material (>1% asbestos) BRL – Below Reporting Limit; Assumed >1% Asbestos	1 – damaged/significantly damaged thermal system insulation ACM 2 – damaged friable surfacing material ACM 3 – significantly damaged friable surfacing material ACM 4 – damaged or significantly damaged friable miscellaneous material ACM 5 – ACM with the potential for damage 6 – ACM with the potential for significant damage 7 – any remaining friable ACM or friable suspected ACM

Materials in **RED** are materials that contain greater than 1% asbestos.
Materials in **BLUE** are assumed to contain >1% asbestos
Materials in **GREEN** contain 1% asbestos or less.
Materials in **BLACK** are none-detected for asbestos.

Sample Information ****See Appendix B for Sample Location Map**		**Material Information**	**Asbestos Content**
HOMOGENEOUS AREA 1	Sample ID: SM1-1 Sample Location: NW bedroom ceiling Sample ID: SM1-2 Sample Location: Bathroom, west wall Sample ID: SM1-3 Sample Location: Sample ID: SM1-4 Sample Location: East entry foyer, north wall Sample ID: SM1-5 Sample Location: Kitchen, south wall	Description: **Smooth Nap Textured Drywall** Classification: Surfacing Material Condition: Good Quantity: ~2,272 ft^2 Friability: Friable Assessment Category: 6 Reason for Assessment: Potential for Contact: High Potential for Vibration: Low Potential for Air Erosion: Low	3% CHRY
HOMOGENEOUS AREA 2	Sample ID: JC1-1 Sample Location: NW bedroom closet, SE corner Sample ID: JC1-2 Sample Location: NE bedroom, SE corner	Description: **Joint Compound Associated w/ SM1** Classification: Miscellaneous Material Condition: Good Quantity: ~2,272 ft^2 Friability: Friable Assessment Category: 6 Reason for Assessment: Potential for Contact: High Potential for Vibration: Low Potential for Air Erosion: Low	3% CHRY

HOMOGENEOUS AREA 3	Sample ID: SM2-1 Sample Location: Living room south wall Sample ID: SM2-2 Sample Location: North hallway ceiling Sample ID: SM2-3 Sample Location: NE bedroom, south wall Sample ID: SM2-4 Sample Location: NE bedroom ceiling Sample ID: SM2-5 Sample Location: Living room south wall	Description: **Sponge Stomp Textured Drywall** Classification: Surfacing Material Condition: Good Quantity: ~1,934 ft^2 Friability: Friable Assessment Category: 6 Reason for Assessment: Potential for Contact: High Potential for Vibration: Low Potential for Air Erosion: Low	2-3% CHRY
HOMOGENEOUS AREA 4	Sample ID: JC2-1 Sample Location: Entry foyer ceiling Sample ID: JC2-2 Sample Location: East hallway, SW corner	Description: **Joint Compound Associated w/ SM2** Classification: Miscellaneous Material Condition: Good Quantity: ~1,934 ft^2 Friability: Friable Assessment Category: 6 Reason for Assessment: Potential for Contact: High Potential for Vibration: Low Potential for Air Erosion: Low	2% CHRY
HOMOGENEOUS AREA 5	Sample ID: SM3-1 Sample Location: Entry foyer, west wall Sample ID: SM3-2 Sample Location: Entry foyer, east wall Sample ID: SM3-3 Sample Location: Entry foyer, north wall	Description: **Heavy Knock Down Textured Drywall** Classification: Surfacing Material Condition: Good Quantity: ~525 ft^2 Friability: Friable Assessment Category: No Category (Non-ACM)	ND
HOMOGENEOUS AREA 6	Sample ID: JC3-1 Sample Location: Entry foyer, south wall Sample ID: JC3-2 Sample Location: Entry foyer, NE corner	Description: **Joint Compound Associated w/ SM3** Classification: Miscellaneous Material Condition: Good Quantity: ~525 ft^2 Friability: Friable Assessment Category: No Category (Non-ACM)	ND

HOMOGENEOUS AREA 7	Sample ID: SM4-1 Sample Location: Living room ceiling, SW side Sample ID: SM4-2 Sample Location: Living room ceiling center Sample ID: SM4-3 Sample Location: Living room ceiling, NW corner	Description: **Swirl Textured Fiberboard** Classification: Surfacing Material Condition: Good Quantity: ~234 ft^2 Friability: Friable Assessment Category: No Category (Non-ACM)	ND
HOMOGENEOUS AREA 8	Sample ID: JC4-1 Sample Location: Living room ceiling, SW side Sample ID: JC4-2 Sample Location: Living room ceiling, NW side	Description: **Joint Compound Associated w/ SM4** Classification: Miscellaneous Material Condition: Good Quantity: ~234 ft^2 Friability: Friable Assessment Category: No Category (Non-ACM)	ND
HOMOGENEOUS AREA 9	Sample ID: CT1-1 Sample Location: Living room ceiling, SW side Sample ID: CT1-2 Sample Location: Living room ceiling, center	Description: **1'x 1' White Ceiling Tiles with Flower and Dot Design** Classification: Miscellaneous Material Condition: Good Quantity: ~234 ft^2 Friability: Friable Assessment Category: No Category (Non-ACM)	ND
HOMOGENEOUS AREA 10	Sample ID: INS1-1 Sample Location: Attic, NW side Sample ID: INS1-2 Sample Location: Attic, north center	Description: **Grey/Beige Cellulose Insulation** Classification: Miscellaneous Material Condition: Good Quantity: ~1,370 ft^2 Friability: Friable Assessment Category: No Category (Non-ACM)	ND
HOMOGENEOUS AREA 11	Sample ID: FT1-1 Sample Location: Kitchen floor, west side Sample ID: FT1-2 Sample Location: Kitchen floor, east side	Description: **12" White/Black Vinyl Floor Tiles and Associated Yellow Adhesive** Classification: Miscellaneous Material Condition: Damaged Quantity: ~170 ft^2 Friability: Non-Friable Assessment Category: No Category (Non-ACM)	ND

HOMOGENEOUS AREA 12	Sample ID: ELCK1-1 Sample Location: West exterior window frame at NW bedroom Sample ID: ELCK1-2 Sample Location: SW exterior window frame at kitchen	Description: **White Exterior Caulking** Classification: Miscellaneous Material Condition: Damaged Quantity: ~16 ft^2 Friability: Non-Friable Assessment Category: No Category (Non-ACM)	ND

4.0 Material Information

A *Homogeneous Area (HA)* means an area of surfacing material, thermal system insulation material, or miscellaneous material that is uniform in color and texture. The asbestos content of the bulk-samples collected within a homogeneous area can be applied to the entire homogenous area, if they conform to the above characteristics and the regulated minimum sample quantities of each type of material have been collected and analyzed. An *Asbestos Containing Material (ACM)* is a material that contains more than 1% asbestos. Any material can be assumed to be an ACM, but not the contrary.

4.1 Material Friability

A material can either be *friable* or *non-friable.* A friable material is one that, when dry, can be pulverized, or reduced to powder by hand pressure, a non-friable material cannot. A non-friable material may become friable if its condition had deteriorated or has been impacted by forces that have rendered it friable.

4.2 Material Classifications

Sampled materials are divided into one of the following three categories:

- *Surfacing Material*: sprayed or troweled onto structural building members
- *Thermal System Insulation (TSI)*: any type of pipe, boiler, tank, or duct insulation
- *Miscellaneous Material*: all other materials not classified in the above two categories

4.3 Material Conditions

Sampled materials are placed into one of the following three categories of conditions:

- *Good*: none to very little visible damage or deterioration
- *Damaged*: the surface is crumbling, blistered, water-stained, gouged, marred, or otherwise abraded over less than one-tenth of the surface if the damage is evenly distributed, or one-quarter if the damage is localized
- *Significantly Damaged*: the surface is crumbling, blistered, water-stained, gouged, marred, or otherwise abraded over greater than one-tenth of the surface if the damage is evenly distributed, or one-quarter if the damage is localized

4.4 Sample Quantities

DS collected at least the minimum number of samples from each homogeneous area necessary to meet all regulatory requirements for the quantity of material to be disturbed in the scope of work as defined by the client. The quantities listed in this report are approximate and on-site verification of the exact quantity of each material is required for permitting, estimating, and billing purposes. The following outlines the minimum

sample quantities required per homogeneous area for a regulatory compliant inspection; however, in the event of a due diligence inspection, these sample minimums may not have been met:

- *Surfacing Materials*: up to 1,000 ft^2 of material requires a minimum of three (3) samples; between 1,000 ft^2 and 5,000 ft^2 of material requires a minimum of five (5) samples; over 5,000 ft^2 of material requires a minimum of seven (7) samples; one (1) sample of each patch
- *Thermal System Insulation (TSI)*: each homogeneous area requires a minimum of three (3) samples; at least one (1) sample must be collected from each patch; and collect enough samples sufficient to adequately assess the material and determine the asbestos content for TSI fittings such as pipe elbows or T's, which a minimum of two (2) samples of each
- *Miscellaneous Materials*: collect enough samples sufficient to determine the asbestos content with a minimum of two (2) samples of each

4.5 Materials Reporting "TRACE" Results

Any sample reporting a "TRACE" amount of asbestos shall be considered to contain greater than 1% asbestos unless it is further analyzed utilizing the point-count method and verified to be less than or equal to 1% asbestos content, and therefore not an ACM. TRACE does not mean it contains less than or equal to 1%.

4.6 Materials Containing 1% Asbestos or Less

Materials containing less than or equal to 1% asbestos are not regulated by the Colorado Department of Public Health and Environment (CDPHE) Regulation 8, Part B – Asbestos. However, all demolition/abatement activities should be performed following the applicable Occupational Safety and Health Administration (OSHA) regulations. This includes, but is not limited to, the appropriate asbestos training for the type of material being removed/disturbed as well as having a properly trained supervisor onsite, using wet removal methods, wearing adequate personal protective equipment (HEPA-filtered particulate respirators), medical surveillance of workers, personal-exposure air monitoring, area air monitoring in occupied buildings, etc. There may also be landfill disposal requirements for these materials, depending on the facility. DS recommends that all demolition/renovation projects involving the disturbance of any amount of asbestos be subjected to post-work visual inspections and a final clearance air testing by a CDPHE-certified Asbestos Air Monitoring Specialist (AMS) after the work has been completed, but before any containments are dismantled, the contractor demobilizes, and the area is reoccupied.

4.7 Overspray

Any surfacing material indicated in this report also includes any associated overspray of that material, e.g., under carpet, above suspended ceilings, on studs and structural members, etc.

5.0 Inspector & Firm Certifications

The inspection detailed within this report was conducted by Mr. Chris Lehman with DS. DS is a CDPHE certified Asbestos Consulting Firm, Registration No. 14912. Mr. Lehman is a CDPHE certified Building Inspector; having certification number 14348 (*see Appendix A for certificates*).

6.0 Inspection, Sampling & Analytical Procedures

6.1 Inspection Procedures

The asbestos inspection detailed in this report was conducted by an Environmental Protection Agency (EPA) and CDPHE certified asbestos Building Inspector. The inspection procedures included identifying and sampling

suspect ACM within the pre-defined areas that were within the scope of work, submitting samples to an accredited laboratory for analysis, classifying the materials and assessing their condition, and compiling a final report detailing the inspection and the analytical results of the bulk-samples.

6.2 Sampling Procedures

Statistically random bulk-samples representative of the suspect ACM of each homogeneous area were collected according to the guidelines published in the Environmental Protection Agency's October 1985 publication, "Asbestos in Building: Simplified Sampling Scheme for Friable Surfacing Materials", commonly known as the "Pink Book."

DS has collected the appropriate number of bulk-samples to meet all regulatory requirements for the classification and quantity of each homogeneous area. All reasonable efforts were made to identify homogeneous areas and to sample or assume suspect materials. Destructive investigation was conducted whenever feasible, and every effort was made to locate and quantify suspect ACM within the scope of work. Any material not identified and sampled in this report shall be assumed to be an ACM or shall be sampled by an EPA-trained and CDPHE-certified inspector and submitted for analysis.

6.3 Analytical Procedures

All asbestos bulk-samples were analyzed by a third party, National Voluntary Laboratory Accreditation Program (NVLAP) accredited laboratory via Polarized Light Microscopy (PLM) for asbestos content per CDPHE Regulation 8 (*see Appendix C for laboratory report*).

7.0 Recommendations

The asbestos inspection detailed in this report did identify ACM(s); therefore, professional abatement activities are required to remove or disturb the above-referenced asbestos-containing materials.

8.0 Asbestos Abatement & Demolition Requirements

If ACM is to be removed or disturbed in a single-family residence, and the total quantity exceeds any of the regulatory trigger levels of 50 linear ft. on pipes, 32 ft^2 on other surfaces, or the volume equivalent of a 55-gallon drum, a CDPHE-certified General Abatement Contractor (GAC) is required to perform the work. The regulatory trigger levels within a commercial building are 260 linear ft. on pipes, 160 ft^2 on other surfaces, or the volume equivalent of a 55-gallon drum. In addition, formal notification to CDPHE prior to the abatement of ACM as well as air monitoring, visual inspections, and final air clearances by a CDPHE-certified Asbestos AMS is required. DS can provide the client or building owner with a proposal for project design, abatement oversight and air monitoring upon request.

CDPHE regulations allow for the demolition of a building that contains certain non-friable asbestos-containing materials, such as caulking, tars, and mastics; however, demolition must be completed without causing the non-friable ACM to be rendered friable. Certain other non-friable materials, such as cementitious siding (Transite) and resilient floor tiles must be abated prior to demolition. DS recommends abating all ACM prior to abatement, regardless of friability. Burning a building with any ACM is prohibited. Operations such as sanding, cutting, crushing, grinding, pneumatic jacking, etc. of ACM are not permitted. Recycling of building materials such as concrete, metal, or wood that are bonded or contaminated with ACM, e.g., glue, caulking, or mastic is also prohibited. If any of the non-friable asbestos containing materials are to be recycled and rendered friable after

demolition (i.e., crushing mastic-coated concrete), these materials must be abated of all ACM prior to shipping offsite for recycling.

OSHA regulations regarding occupational exposure during demolition activities is still mandatory. OSHA 29 CFR 1926.1101 requires that workers performing construction-related activities be protected from asbestos fibers more than the permissible exposure limit of 0.1 f/cc of air. Contractors must comply with applicable provisions of OSHA 29 CFR 1926.1101 during demolition and renovation activities. These OSHA provisions include, but are not limited to, PPE and respirators, personnel training, personal-exposure air monitoring, employee medical surveillance, wet removal methods, signage for regulated areas, etc.

9.0 Major Asbestos Spills

If ACM is significantly damaged and the total quantity exceeds the regulatory trigger levels, the area is deemed a "Major Asbestos Spill." The area is consequently subject to the requirements in Reg. 8, Section III.T.2. – *Major Asbestos Spills*. Unless the entire facility is to be treated as a major asbestos spill, a Colorado-certified Air Monitoring Specialist (AMS) must determine the extent of the spill area. This may be done using visual examination, air samples, micro-vacuum dust samples, wipe samples or a combination thereof. If visible dust or debris is observed, directly related to or resulting from the known or assumed ACM which created the major asbestos spill, areas where it is observed must be included in the abatement of the spill. Samples must be collected and analyzed quantitatively by Transmission Electron Microscopy (TEM.)
The General Abatement Contractor (GAC) selected to perform the cleanup of the spill must:

- Submit notification in accordance with subsection III.E. (Notifications) or subsection III.G. (Permits), whichever is applicable to the Division for approval.
- Using certified Workers and Supervisors, in accordance with Section II. (Certification Requirements), construct a containment in accordance with the requirements of the regulation.
- HEPA vacuum then steam clean all carpets, drapes upholstery and other non-clothing fabrics in the contaminated area or discard these materials in accordance with subsection III.R. (Waste Handling)
- Launder or discard all contaminated clothing in accordance with subsection III.R. (Waste Handling)
- HEPA vacuum or wet wipe with clean amended water all hard surfaces in the contaminated area.
- Discard all waste in accordance with subsection III.R. (Waste Handling)

All persons must comply with any other measures, provided in writing by the Division, which are deemed necessary to protect public health. Following completion of Sections III.T.2.d.(i) through III.T.2.e., the AMS must comply with air monitoring requirements as described in Section III.P. (Clearing Abatement Projects); air samples must be collected aggressively as described in 40 C.F.R. Part 763, Appendix A to Subpart E (EPA 2010), except that the air stream of the leaf blower must not be directed at any friable ACM that remains in the area. Gross removal of additional ACM may not be conducted under Section III.T.2. Any remaining gross removal of ACM must be abated in accordance with Section III.H. (Abatement Sequence). If additional ACM is to be removed, the final air sampling required in Section III.T.2.f. is not required to be conducted until after the additional removal is completed.

10.0 Project Design & Project Manager Requirements

DS can provide an Asbestos Abatement Project Design as well as fulfill the Colorado Asbestos Abatement Project Manager requirements for any asbestos abatement project, as applicable below.

Project Design
An abatement *Project Design* is an accurate and detailed scope of work, which includes project specifications and procedures, containment design/equipment placement, and descriptions of engineering controls and work practices for an asbestos abatement project or response action that is required by CDPHE Regulation Number 8, Part B - Asbestos (Reg. 8) on large asbestos abatement projects. Prior to the start of any asbestos abatement project in a non-school building, where the amount of asbestos-containing material (ACM) to be removed or disturbed exceeds 1,000 linear feet on pipes, or 3,000 square feet on surfaces, or in a school building in which the amount of friable ACM to be abated exceeds 3 linear feet on pipes, or 3 square feet on surfaces, a written Project Design must be developed by a State of Colorado certified Project Designer in accordance with subsection IV.G.7 of Regulation 8. A signed copy shall be posted on-site prior to commencing any abatement activities, shall be always available on-site, and shall remain onsite until final air clearances have been completed by a State of Colorado-certified Air Monitoring Specialist (AMS).

Project Manager
A *Project Manager* shall be used on all asbestos abatement projects in which the amount of friable asbestos-containing material to be abated exceeds 1,000 linear feet on pipes, or 3,000 square feet on other surfaces per CDPHE Regulation Number 8, Part B – Section III.B.6. An asbestos Project Manager on an abatement project shall be responsible for assessing that the project is conducted in accordance with Regulation 8, assessing that the Project Design is followed, assessing that the abatement project is cleared in accordance with Regulation 8, assessing that the asbestos waste generated on the project is properly manifested and disposed of in accordance with Regulation 8, and communicating these assessments to the building owner or GAC.

The GAC shall notify the building owner during the bid process as to whether a Project Manager is required. Project Managers shall be independent of the asbestos abatement contractor and work strictly on behalf of the building owner to the extent feasible unless the abatement is being performed in-house. Project Managers must sign the original copy of the abatement permit for the permit to be valid, and before any abatement can take place.

11.0 Disclaimer & Limitations

The activities outlined in this report were conducted in a manner consistent with a level of care and expertise exercised by members of the environmental consulting and industrial hygiene profession. All activities were performed in accordance with all applicable federal, state, and local regulations as well as generally accepted standards and professional practice. No warranty is either expressed or implied. DS assumes no responsibility or liability for error in public information utilized, statements from sources other than DS, or developments resulting from situations outside the scope of work for this project.

The details provided within this report outline the inspection activities on the date(s) indicated and should not be relied upon to represent conditions later. The laboratory results contained in this report apply specifically to the materials in which bulk-samples were collected. The results do not include or apply to any other materials within the structure that were not sampled but may contain asbestos; including materials that may be hidden or inaccessible. Additional inspection and bulk-sampling activities by a certified inspector would be required to determine whether any other materials contain asbestos.

This report has been prepared on behalf of and exclusively for use by the DS's client, with specific application to their project as discussed in the scope of work. The information contained in this report is intended as supplementary material for abatement design and is not to be used as the sole means to develop the scope of

abatement activities, bidding, or billing purposes. Contractors or consultants reviewing this report must draw their own conclusions regarding further investigation or remediation deemed necessary. DS can provide a full scope of work for abatement upon request. DS does not warrant the work of regulatory agencies, laboratories or other third parties supplying information which may have been used in the preparation of this report.

12.0 Copyright Notice

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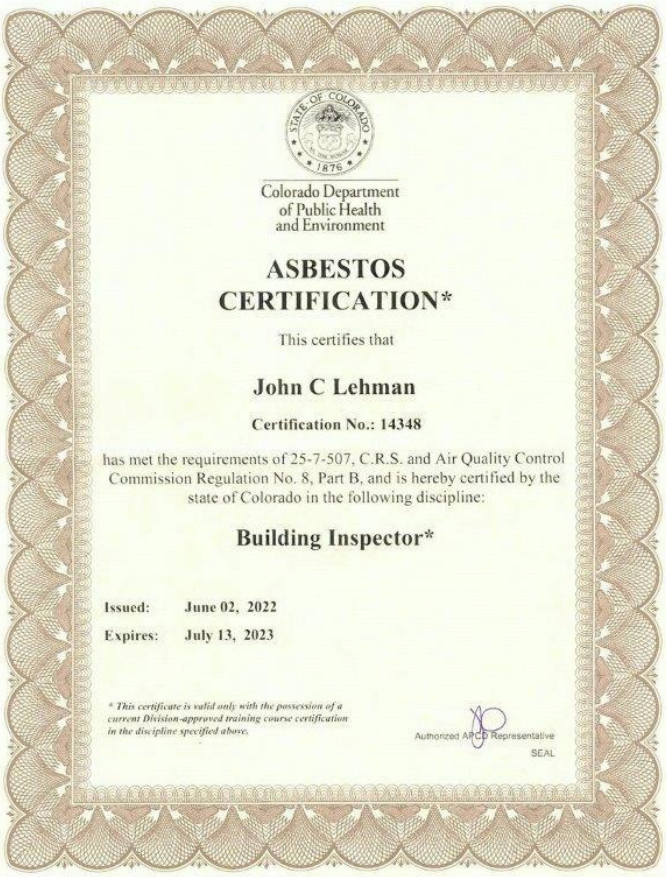
APPENDIX A: CERTIFICATIONS



Colorado Department
of Public Health
and Environment

ASBESTOS CERTIFICATION*

This certifies that

John C Lehman

Certification No.: 14348

has met the requirements of 25-7-507, C.R.S. and Air Quality Control Commission Regulation No. 8, Part B, and is hereby certified by the state of Colorado in the following discipline:

Building Inspector*

Issued: **June 02, 2022**

Expires: **July 13, 2023**

** This certificate is valid only with the possession of a current Division-approved training course certification in the discipline specified above.*

Authorized APCD Representative

SEAL

Inspector Certification: 14348

Colorado Department
of Public Health
and Environment

ASBESTOS CONSULTING FIRM

This certifies that

DS Environmental Consulting, Inc.

Registration No.: ACF - 14912

has met the registration requirements of 25-7-507, C.R.S. and the Air Quality Control Commission Regulation No. 8, Part B, and is hereby authorized to perform asbestos consulting activities as required under Regulation No 8, Part B, in the state of Colorado.

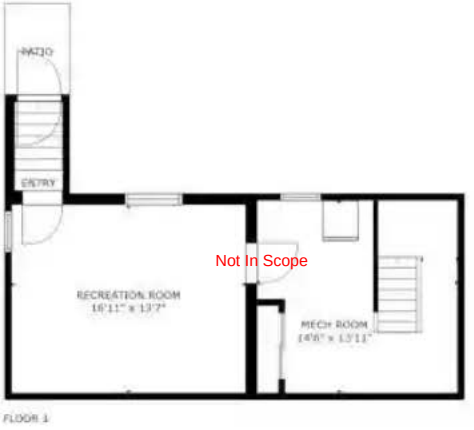
Issued: January 24, 2022
Expires: January 30, 2023

Authorized APCD Representative

SEAL

Firm Certification: 14912

APPENDIX B: SAMPLE LOCATIONS AND ACM HOMOGENEOUS AREAS



KEY:

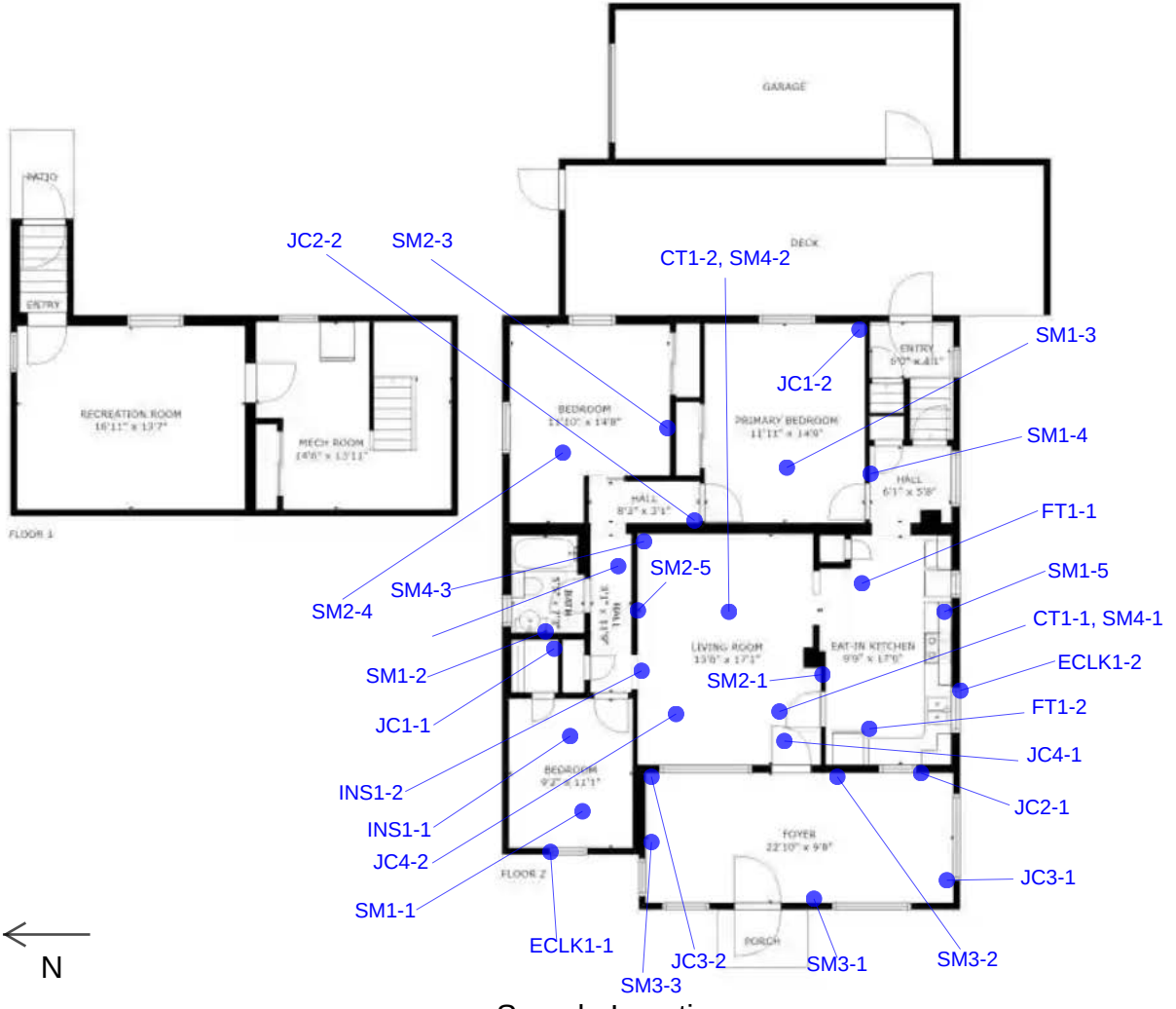
Drywall Walls and Ceilings SM1/JC1 SM2/JC2

Drywall Ceilings SM1/JC1 SM2/JC2

Drywall Walls SM2/JC2

N

ACM Locations



Sample Locations

APPENDIX C: LABORATORY REPORT

Aerobiology Laboratory Associates, Incorporated
Expertise Since 1997

780 Simms Street
Suite 104
Golden, CO, 80401
303.232.3746
www.aerobiology.net

Certificate of Analysis

NVLAP TESTING
NVLAP Lab Code 200860-0

Client Name	DS Environmental Consulting	Date Collected:	6/16/2022
Street Address	7555 W. 10th Ave, Suite A	Date Received:	6/17/2022
City, State ZIP	Lakewood, CO 80214	Date Analyzed:	6/23/2022
Attn:	Chris Lehman	Date Reported:	6/24/2022
Client Project Name:	275 Holbrook St, Erie, CO	Project ID:	22023167

Test Requested: **3002, Asbestos in Bulk Samples**
Method: EPA 600/R-93/116: Method for Asbestos in Bulk Building Materials, EPA -- 40 CFR Appendix E to Subpart E of Part 763, Interim Method for Asbestos in Bulk Insulation Samples

Sample Identification		Layer Percentage	Physical Description of Sample/Layer	Asbestos Detected	Asbestos Percentage	Non-Asbestos Fiber Percentage	Non-Fibrous Material Percentage	Matrix Material Composition	Homo-geneous (Y/N)
Client	Lab Sample Number								
SM1-1	22023167-1A	25	White Texture with White Paint	ND			100	C	N
	22023167-1B	75	Gray/Tan Drywall	ND		30 CELL	70	G	N
SM1-2	22023167-2A	40	White Texture with Red/Multicolored Paint	ND			100	C	N
	22023167-2B	60	Off-White/Tan Drywall	ND		45 CELL	55	G	N
SM1-3	**22023167-3A**	**50**	**White Compound with White Paint**	**CHRY**	**3**		**97**	**G**	**N**
	22023167-3B	50	Tan/Off-White Drywall	ND		60 CELL	40	G	N
SM1-4	**22023167-4A**	**25**	**Off-White Texture with Off-White/Multicolored Paint**	**CHRY**	**3**		**97**	**C**	**N**
	22023167-4B	75	Off-White/Tan Drywall	ND		30 CELL	70	G	N
SM1-5	22023167-5A	15	White Compound with Off-White/Tan Paint	ND			100	G	N
	22023167-5B	85	Tan/White Drywall	ND		65 CELL	35	G	N

Thomas Harbour
Laboratory Analyst

Shannon Whitmore
Asbestos Lab Supervisor

AC = Actinolite	AH = Animal Hair	B = Binder	Q = Quartz
AM = Amosite	CELL = Cellulose	C = Calcite	T = Tar
AN = Anthophyllite	FG = Fibrous Glass	D = Diatoms	V = Vermiculite
CHRY = Chrysotile	MW = Mineral Wool	G = Gypsum	
CR = Crocidolite	OT = Other	M = Mica	
TRM = Tremolite	SYN = Synthetic	OR = Organic	
Tr = Trace	TL = Talc	OP = Opaques	
ND = None Detected	W = Wollastonite	P = Perlite	

Aerobiology Laboratory Associates, Incorporated
Expertise Since 1997

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Certificate of Analysis

NVLAP TESTING
NVLAP Lab Code 200860-0

Client Name	DS Environmental Consulting	Date Collected:	6/16/2022
Street Address	7555 W. 10th Ave, Suite A	Date Received:	6/17/2022
City, State ZIP	Lakewood, CO 80214	Date Analyzed:	6/23/2022
Attn:	Chris Lehman	Date Reported:	6/24/2022
Client Project Name:	275 Holbrook St, Erie, CO	Project ID:	22023167

Test Requested: **3002, Asbestos in Bulk Samples**
Method: EPA 600/R-93/116: Method for Asbestos in Bulk Building Materials, EPA -- 40 CFR Appendix E to Subpart E of Part 763, Interim Method for Asbestos in Bulk Insulation Samples

Sample Identification		Layer Percentage	Physical Description of Sample/Layer	Asbestos Detected	Asbestos Percentage	Non-Asbestos Fiber Percentage	Non-Fibrous Material Percentage	Matrix Material Composition	Homo-geneous (Y/N)
Client	Lab Sample Number								
JC1-1	22023167-6A	7	White Compound with White Paint	ND			100	C,G	N
	22023167-6B	5	White Tape	ND		95 CELL	5		N
	22023167-6C	8	White Joint Compound	ND			100	C,G	Y
	22023167-6D	80	Off-White/Tan Drywall	ND		15 CELL,FG	85	G,M	N
JC1-2	**22023167-7A**	**32**	**White Compound with Light Purple/Multicolored Paint**	**CHRY**	**3**		**97**	**G**	**N**
	22023167-7B	10	White Tape	ND		95 CELL	5		N
	22023167-7C	**25**	**White Joint Compound**	**CHRY**	**3**		**97**	**G**	**Y**
	22023167-7D	33	Off-White/Tan Drywall	ND		15 CELL	85	G	N
SM2-1	**22023167-8A**	**45**	**Off-White Texture with Off-White/Multicolored Paint**	**CHRY**	**2**		**98**	**C**	**N**
	22023167-8B	55	Off-White/Tan Drywall	ND		40 CELL	60	G	N

Thomas Harbour
Laboratory Analyst

Shannon Whitmore
Asbestos Lab Supervisor

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AN = Anthophyllite	FG = Fibrous Glass	D = Diatoms	V = Vermiculite
CHRY = Chrysotile	MW = Mineral Wool	G = Gypsum	
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Sample Identification		Layer Percentage	Physical Description of Sample/Layer	Asbestos Detected	Asbestos Percentage	Non-Asbestos Fiber Percentage	Non-Fibrous Material Percentage	Matrix Material Composition	Homo-geneous (Y/N)
Client	Lab Sample Number								
SM2-2	22023167-9A	30	Gray Perlitic Texture with White Paint	ND			100	C,P	N
	22023167-9B	**8**	**Off-White Compound**	**CHRY**	**2**		**98**	**C**	**N**
	22023167-9C	62	Off-White/Tan Drywall	ND		30 CELL	70	G	N
SM2-3	**22023167-10A**	**45**	**Off-White Texture with Off-White/Multicolored Paint**	**CHRY**	**2**		**98**	**C**	**N**
	22023167-10B	55	Off-White/Tan Drywall	ND		45 CELL	55	G	N
JC2-1	22023167-11A	30	Off-White Texture with White Paint	ND			100	C	N
	22023167-11B	20	Off-White Texture with Off-White Paint	ND			100	C,G	N
	22023167-11C	10	White Tape	ND		95 CELL	5		N
	22023167-11D	15	White Joint Compound	ND			100	G	Y
	22023167-11E	25	Off-White/Tan Drywall	ND		15 CELL	85	G	N

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Client Name	DS Environmental Consulting	NVLAP TESTING	Date Collected:	6/16/2022
Street Address	7555 W. 10th Ave, Suite A	NVLAP Lab Code 200860-0	Date Received:	6/17/2022
City, State ZIP	Lakewood, CO 80214		Date Analyzed:	6/23/2022
Attn:	Chris Lehman		Date Reported:	6/24/2022
Client Project Name:	275 Holbrook St, Erie, CO		Project ID:	22023167

Test Requested: **3002, Asbestos in Bulk Samples**
Method: EPA 600/R-93/116: Method for Asbestos in Bulk Building Materials, EPA -- 40 CFR Appendix E to Subpart E of Part 763, Interim Method for Asbestos in Bulk Insulation Samples

Sample Identification		Layer Percentage	Physical Description of Sample/Layer	Asbestos Detected	Asbestos Percentage	Non-Asbestos Fiber Percentage	Non-Fibrous Material Percentage	Matrix Material Composition	Homo-geneous (Y/N)
Client	Lab Sample Number								
JC2-2	**22023167-12A**	**7**	**Off-White Compound with White/Yellow Paint**	**CHRY**	**2**		**98**	**C**	**N**
	22023167-12B	5	White Tape	ND		95 CELL	5		N
	22023167-12C	**9**	**Off-White Joint Compound**	**CHRY**	**2**		**98**	**C**	**Y**
	22023167-12D	79	Off-White/Tan Drywall	ND		15 CELL	85	G	N
SM3-1	22023167-13A	50	Off-White Texture with White Paint	ND			100	C	N
	22023167-13B	25	Off-White Texture with Off-White Paint	ND			100	G	N
	22023167-13C	25	Tan/Off-White Drywall	ND		70 CELL	30	G	N
SM3-2	22023167-14A	55	Off-White Texture with White Paint	ND			100	C	N
	22023167-14B	10	Off-White Texture with Off-White Paint	ND			100	G	N
	22023167-14C	35	Off-White/Tan Drywall	ND		35 CELL	65	G	N

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City, State ZIP	Lakewood, CO 80214	Date Analyzed:	6/23/2022
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Sample Identification		Layer Percentage	Physical Description of Sample/Layer	Asbestos Detected	Asbestos Percentage	Non-Asbestos Fiber Percentage	Non-Fibrous Material Percentage	Matrix Material Composition	Homo-geneous (Y/N)
Client	Lab Sample Number								
SM3-3	22023167-15A	50	Off-White Texture with White Paint	ND			100	C	N
	22023167-15B	10	Off-White Texture with Off-White Paint	ND			100	G	N
	22023167-15C	40	Off-White/Tan Drywall	ND		25 CELL	75	G	N
JC3-1	22023167-16A	15	Off-White Texture with White Paint	ND			100	C	N
	22023167-16B	10	Off-White Texture with Off-White Paint	ND			100	G	N
	22023167-16C	8	White Tape	ND		95 CELL	5		N
	22023167-16D	7	Off-White Joint Compound	ND			100	G	Y
	22023167-16E	60	Off-White/Tan Drywall	ND		15 CELL	85	G	N
JC3-2	22023167-17A	35	Off-White Texture with White Paint	ND			100	C	N
	22023167-17B	20	Off-White Texture with Off-White Paint	ND			100	G	N

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Sample Identification		Layer Percentage	Physical Description of Sample/Layer	Asbestos Detected	Asbestos Percentage	Non-Asbestos Fiber Percentage	Non-Fibrous Material Percentage	Matrix Material Composition	Homo-geneous (Y/N)
Client	Lab Sample Number								
JC3-2	22023167-17C	45	Off-White/Tan Drywall	ND		20 CELL	80	G	N
SM4-1	22023167-18A	20	White Texture with Pink/Multilayered Paint	ND		2 TL	98	C,M	N
	22023167-18B	80	Tan Fiberboard	ND		85 CELL	15		Y
SM4-2	22023167-19A	20	White Texture with Light Tan Paint	ND		2 TL	98	C,M	N
	22023167-19B	80	Tan Fiberboard	ND		85 CELL	15		Y
SM4-3	22023167-20A	25	White Texture with Gray Paint	ND		2 TL	98	C,M	N
	22023167-20B	75	Tan Fiberboard	ND		85 CELL	15		Y
JC4-1	22023167-21A	6	Light Pink Compound with Light Tan Paint	ND		Tr TL	100	C,M	N
	22023167-21B	4	White Woven Material	ND		90 CELL	10		Y
	22023167-21C	50	White/Light Tan Compound	ND		1 TL	99	C,M	N

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City, State ZIP	Lakewood, CO 80214	Date Analyzed:	6/23/2022
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Sample Identification		Layer Percentage	Physical Description of Sample/Layer	Asbestos Detected	Asbestos Percentage	Non-Asbestos Fiber Percentage	Non-Fibrous Material Percentage	Matrix Material Composition	Homo-geneous (Y/N)
Client	Lab Sample Number								
JC4-1	22023167-21D	40	Tan Fiberboard	ND		80 CELL	20		Y
JC4-2	22023167-22A	10	Light Pink Compound with Light Blue Paint	ND		Tr TL	100	C,M	N
	22023167-22B	15	White Woven Material	ND		90 CELL	10		Y
	22023167-22C	15	White/Light Tan Compound	ND		1 TL	99	C,M	N
	22023167-22D	60	Tan Fiberboard	ND		80 CELL	20		Y
CT1-1	22023167-23A	7	White Paint with White Texture	ND			100	B	N
	22023167-23B	93	Tan Fiberboard Ceiling Tile	ND		85 CELL	15		Y
CT1-2	22023167-24A	8	White Textured Paint	ND		2 CELL	98	B	Y
	22023167-24B	92	Tan Fiberboard	ND		85 CELL	15		Y
INS1-1	22023167-25	100	Gray Fibrous Insulation	ND		85 FG,CELL	15		Y

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Client	Lab Sample Number								
INS1-2	22023167-26	100	Gray Fibrous Insulation	ND		85 FG,CELL	15		Y
FT1-1	22023167-27A	98	White Vinyl Flooring with Black Paint	ND			100	B,C	N
	22023167-27B	2	Light Tan Mastic	ND			100	B,C	Y
FT1-2	22023167-28A	99	Black Vinyl Flooring	ND			100	B,C	N
	22023167-28B	1	Light Tan Mastic	ND			100	B	Y
ECLK1-1	22023167-29	100	White Caulking with Blue Paint	ND			100	B,C	Y
ECLK1-2	22023167-30	100	White Caulking with Blue Paint	ND			100	B,C	Y
SM2-4 (NOT ON ORIGINAL COC)	22023167-31A	3	White Compound	ND			100	C,M	Y
	22023167-31B	20	White/Multilayered Paint	ND			100	B	Y
	22023167-31C	**20**	**White Compound**	**CHRY**	**3**		**97**	**C,M**	**Y**

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Sample Identification		Layer Percentage	Physical Description of Sample/Layer	Asbestos Detected	Asbestos Percentage	Non-Asbestos Fiber Percentage	Non-Fibrous Material Percentage	Matrix Material Composition	Homo-geneous (Y/N)
Client	Lab Sample Number								
SM2-4 (NOT ON ORIGINAL COC)	22023167-31D	57	Light Pink/Tan Drywall	ND		65 CELL	35	G	N
SM2-5 (NOT ON ORIGINAL COC	22023167-32A	15	White Paint	ND			100	B	Y
	22023167-32B	**25**	**White Texture**	**CHRY**	**3**		**97**	**C,M**	**Y**
	22023167-32C	60	Light Pink/Tan Drywall	ND		65 CELL	35	G	N

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Client Name	DS Environmental Consulting	NVLAP TESTING	Date Collected:	6/16/2022
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City, State ZIP	Lakewood, CO 80214		Date Analyzed:	6/23/2022
Attn:	Chris Lehman		Date Reported:	6/24/2022
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General Notes

- **ND** indicates no asbestos was detected; the method detection limit is 1 %.
- **Trace** or "< 1" indicates asbestos was identified in the sample, but the concentration is less than 1% and cannot be quantified without point counting.
- Samples identified as inhomogeneous (more than one layer) are separated into individual layers, and each layer is analyzed and reported separately.
- All regulated asbestos minerals (i.e. chrysotile, amosite, crocidolite, anthophyllite, tremolite, and actinolite) were sought in every layer of each sample, but only those asbestos minerals detected are listed. Amosite is the common name for the asbestiform variety of the mineral grunerite. Crocidolite is the common name used for the asbestiform variety of the mineral riebeckite.
- Tile, vinyl, foam, plastic, and fine powder samples may contain asbestos fibers of such small diameter (< 0.25 microns in diameter) that these fibers cannot be detected by PLM. For such samples, more sensitive analytical methods (e.g. TEM, SEM, and XRD) are recommended if greater certainty about asbestos content is required. Semi-quantitative bulk TEM floor tile analysis is accepted under NESHAP regulations.
- These results are submitted pursuant to Aerobiology Laboratory Associates, Inc.'s current terms and conditions of sale, including the company's standard warranty and limitation of liability provisions. No responsibility or liability is assumed for the manner in which the results are used or interpreted.
- Unless notified in writing to return the samples covered by this report, Aerobiology Laboratory Associates, Inc. will store the samples for a minimum period of thirty (30) days before discarding. A shipping and handling charge will be assessed for the return of any samples.
- Aerobiology does not guarantee the results of tape lifts, microvacs, wipe, and/or debris samples. Accurate analysis cannot be performed due to particle size, media used, and/or amount of material given. Analysis of these materials should be preformed by a TEM. ***A result of ND does not indicate that the sample area does not contain asbestos. It means the analyst could not identify asbestos in the specific sample for the reasons listed above.***
- "When joint compound and/or tape is applied to a wallboard it becomes an integral part of the wallboard and in effect becomes one material forming a wall system." EPA 40 CFR Part 61 Aerobiology cannot distunguish joint compound from the same material used as skim coat. Therefore, it is very important that individuals collecting the samples clearly describe the sample composition so Aerobiology knows that the drywall system can be composited. If only joint sampling areas show layers with >1% asbestos, then material is joint compound. If samples from both joint sampling area and non-joint areas show layers with >1% asbestos, then the material should be considered "skim coat" or add-on material.

Notes Required by NVLAP

- This report must not be used by the client to claim product certification, approval, or endorsement by NVLAP, NIST, or any agency of the Federal Government.
- This test report relates only to the items tested or calibrated.
- This report is not valid unless it bears the name of a NVLAP-approved signatory.
- Any reproduction of this document must include the entire document in order for the report to be valid.

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Lab Use: 22023167

ELITE AZ, CA, CO, FL, GA, VA, NJ

NVLAP AZ, CA, CO, VA

AZ, CA, CO, FL GA, NJ, VA

Aerobiology Client	DS Environmental Consulting, Inc.		
Field Contact	Chris Lehman	Collected By/Date:	Relinquished By/Date: 6/17/22
Reporting Address		Relinquished By/Date:	Received By/Date: 6/17/22
Billing Address	7555 W. 10th Ave., Lakewood CO	Sampler Type	Andersen___ SAS___ SampleAire___ AeroTrap___ Other___ BioCulture___
Phone/Fax	720-369-6609	PO#/Job#:	
Reporting Email (s)	chris@dsconsultinginc.com & DS Distribution Group	Project Name:	275 Holbrook St, Eric, CO
Routine ☒	24 Hour ☐ Same Day ☐ 4 Hour ☐ 2 Hour ☐	Notes:	
SAMPLING LOCATION ZIP CODE		CC Info:	

	Sample No.	Test Code	Sample Location	Total Volume/Area
1	SM1-1	3002		
2	1-2			
3	1-3			
4	1-4			
5	1-5			
6	JC1-1			
7	1-2			
8	SM2-1			
9	2-2			
10	2-3			
11	JC2-1			
12	2-2			
13	SM3-1			
14	3-2			
15	3-3			

1054	Direct, Non-viable Spore Trap	1015	Culture - WATER Legionella
1051	Direct, Qualitative- **Swab/Tape**	1017	Culture - SWAB Legionella
1050	Direct, Qualit		coli/total coliforms
1005	AIR Culture -		liforms
1030	AIR Culture -		en (E. coli/Entero/fecal coliforms)
1006	SWAB Cultu		ic Plate Count
1031	SWAB Cultu		unt
1008	BULK Cultur		alysis
1033	BULK Cultur		characterization
1007	WATER Cult		nalysis

LAB USE ONLY

A: TH ✓ 1-17 (+) MS 10-32 (+) 6/23/22 (+)

V:

Q: SW 6/24/2022

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Revision 12

Aerobiology Laboratory Associates, Incorporated
Expertise Since 1997

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ELITE AZ, CA, CO, FL, GA, VA, NJ

NVLAP AZ, CA, CO, VA

AZ, CA, CO, FL GA, NJ, VA

Aerobiology Client	DS Environmental Consulting, Inc.		
Field Contact	Chris Lehman	Collected By/Date:	Relinquished By/Date:
Phone/Fax	720-369-6609	PO#/Job#:	
Email	chris@dsconsultinginc.com	Project Name:	

	Sample No.	Test Code	Sample Location	Total Volume/Area
16	JC34	3002		
17	13-2			
18	SM4-1			
19	4-2			
20	4-3			
21	JC4-1			
22	4-2			
23	CT1-1			
24	1-2			
25	INS1-1			
26	1-2			
27	FT1-1			
28	1-2			
29	ECLK1-1			
30	1-2			
31	SM2-4		Not on original COC	
32	SM2-5		Not on original COC	
33				
34				
35				
36				
37				
38				
39				
40				